SECTION 13930-A
SPRINKLER SYSTEMS

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Related Sections: The following Sections contain requirements that relate to this Section:
   1. Section 09900 – Painting Equipment and Piping.

C. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements

D. In the event of conflict regarding the requirements between this Section and any other section, the provisions of this Section shall govern.

1.2 SUMMARY

A. Section includes automatic sprinkler system components, such as piping, valves, sprinkler heads, and other devices required to provide a fire protection system for the hazard classification specified.

1.3 QUALITY ASSURANCE

A. Provide sprinkler equipment and installation in accordance with recommendations of FM and as approved and required by ROC local authorities and code.

B. Equipment and installation shall meet requirements of Uniform Building Code and Uniform Fire Code 1997 (UBC, UFC) including UBC Standard 9-1 and FM (Factory Mutual) Data Sheets 2-8N, 7-7 (17-12), and 3-10; 13 - Standard for the Installation of Sprinkler Systems, 16 - Foam-Water Sprinkler and Spray Systems, except as modified by these Specifications.

C. Design Criteria:
   1. Refer to the Drawings and requirements contained in this Section for fire protection system design criteria.

2. The system shall be hydraulically designed and subsequently installed in accordance with the hazard classification as indicated. Calculations shall be submitted for approval.

D. Seismic Protection Design: Design and install fire protection system piping sway bracing per NFPA 13 and FM Data Sheet 2-8 based on a horizontal force factor of 75 percent of the weight of water filled pipes.

1.4 APPROVALS

A. The following fire protection components are required to be FM approved:

Pressure switches	Flow switches	Foam sprinklers
Flow meters	Sprinklers	
Check valves	Spray nozzles	Flexible couplings
Alarm check valves	Flexible hose	
OS&Y gate valves 4" and above	Deluge valve – solenoid or pneumatic	Smoke/heat detectors Flame detectors
Foam tank		Air sampling detectors
Proportional mixer		

B. The following must have the approval of UL, FM, or BS:

Alarm panels	Firestop	Tamper switches

C. The following are not required to be FM approved:

Indoor hydrants	Fire extinguishers	Outdoor hydrants Hose valves
Seismic bracing & hangers		

1.5 SYSTEM DESCRIPTION

A. Wet-Pipe Sprinkler System: System with automatic sprinklers attached to piping system containing water and connected to water supply so that water discharges immediately from sprinklers when they are opened by fire.

B. Deluge Sprinkler System: System with open sprinklers attached to piping system connected to water supply through deluge valve. Valve is opened by operation of fire detection system in same areas as sprinklers. When valve opens, water flows into piping system and discharges from attached sprinklers. Design systems serving cooling towers to meet requirements of FM Data Sheet 1-6.

C. Sprinkler System Protection Limits: All spaces within areas indicated. Include closets, toilet and locker room areas, each landing of each stair, and special applications areas.

1.6 SUBMITTALS

A. Submit shop drawings in accordance with Division 1.
   1. Submit shop drawings of entire sprinkler system.
   2. Submit hydraulic calculations.
   3. Submit system test verification.

PART 2 - PRODUCTS

2.1 COMPONENTS

A. Provide components listed in the FM Approval Guide.

2.2 ACCEPTABLE MANUFACTURERS

A. Sprinkler Heads and Accessories:
   1. Automatic Sprinkler
   2. ITT Grinnell
   3. Victaulic
   4. Viking

B. Grooved Fittings:
   1. ITT Grinnell
   2. Gustin-Bacon
   3. Victaulic

2.3 SPRINKLERS

A. Acceptable Manufacturers: from approved list of manufacturers of UL-listed and FM-approved products for fire sprinkler systems.

B. General Description:

1. Orifice Diameter and Temperature Rating: Refer to room schedules on the Drawings.
2. Type: quartzoid bulb, chrome plated, pendent or upright, bronze in exposed areas.
3. Guards: listed type where subject to mechanical damage or below 2.2 m height above floor.

C. Cleanroom Ceilings:
1. Acceptable Manufacturer: FlexHead Industries.
2. Description: Quick-response, quartzoid bulb, 25.4 mm stainless steel flexible metal hose assembly.

D. Ductwork:
1. Prefabricated Assembly:
    a. Acceptable Manufacturer: FlexHead Industries.
    b. Description:
        1) Flexible Assembly: 25.4 mm seamless stainless steel flexible metal hose with 25.4 mm by 12.7 mm stainless steel bell reducer, flange, EPDM gasket, and base-mounting block compatible with duct material.
        2) Sprinkler Type: bronze, fusible link pendent, 100 degrees C.
        3) Orifice: 9.5 mm for ductwork 60 cm and smaller in diameter and 12.7 mm for ductwork larger than 60 cm in diameter.
    c. Corrosion Protection: single-layer, factory-applied layer of beeswax coating, double bagged in 0.1 mm thick polyethylene.

E. Other Finished Ceiling Areas - Type: recessed pendent, chrome finish, quartzoid bulb with chrome-plated canopy.

2.4 SPRINKLERS – APPLICATION REQUIREMENTS

A. Place sprinklers in upright or pendent position as required, with the deflector parallel to the ceiling. Maintain clearances between the deflectors or other obstructions in accordance with FM Data Sheet 2-8N and NFPA 13.

B. Fab, cleanroom, and subfab areas:
1. For areas with cleanroom ceiling, use semi-recessed heads. Flexible riser nipple assemblies used to attach the branch pipe and sprinkler head to the cleanroom ceiling shall use FM approved components.
2. Additional friction loss for the flexible nipples shall be included in the hydraulic calculations.

C. Pass Throughs: Standard 10 mm orifice quick response heads.

D. Exposed Areas: Standard upright type with brass finish.

E. Sidewall Applications: Provide sidewall type with chrome-plated finish and escutcheon.

F. Acid Scrubbers: Provide sprinklers with a minimum flow rate of 20 gpm per head at the outlet and inlet transitions and at the top panel of each of the acid exhaust system scrubbers. The sprinklers shall be FM approved wax coated and bagged for corrosion protection. Each sprinkler shall be connected to the branch pipe by an FM approved flexible nipple for ease of sprinkler removal for inspection.

G. Cooling Towers: Where required by FM, cooling towers shall be protected by automatic deluge systems, with separate piping and activation system per cell. The systems shall be sized to provide deluge to two cells simultaneously. The sprinkler protection shall be installed under the fan deck, including the fan opening, designed to provide a minimum density of 0.50 gpm/ft^2. All piping downstream from the deluge valve shall be stainless steel.

H. Drum Storage Rooms: If drums are stored in single- or double-row racks, one level of in-rack sprinkler protection shall also be provided beneath each level of storage utilizing 74 degrees C heads spaced every 2.75 m and hydraulically designed to provide 112 l/min per head while flowing the most remote 6 heads (8 heads if only one level is provided).

I. Gas Cabinets: Sprinklers shall be provided in all flammable gas cabinets, hydraulically designed to provide 75 l/min per head with 5 heads flowing, utilizing 57 degrees C, 9.5 mm orifice heads.

J. Bulk Flammable and Pyrophoric Gas (including silane) Storage: A fixed spray system shall be provided over the flammable or pyrophoric gas tanks, pumps, regulator systems and control panel areas for cooling in a fire situation. The systems shall be hydraulically designed to provide 12 $l/min\text{-}m^2$ over the external tank surface area. The systems shall be activated by FM approved optical flame detection, interlocked to shut-off the gas and sound alarm at the emergency response center.

K. Provide head guards when the sprinklers are subject to damage or are located less than 2.2 m above the floor.

2.5 SPRINKLER RISER

A. General: Provide automatic sprinkler dry pipe or deluge valve for systems as indicated. Provide flow switches for wet systems. Provide pressure switches for dry systems. Provide all required pressure gauges, inspectors test and drain valves, sight glasses, pipe and fittings.

2.6 VALVES AND TAMPER SWITCHES

A. Valves: Threaded through 2 inches, grooved or shouldered in sizes 2-1/2 inches and larger to match pipe and fittings specified. Provide enamel on metal identification sign on all valves. Provide all valves except test and drain valves with steel lock and chains with all locks keyed alike.

B. Gate Valves: OS&Y, 175 psi WWP. Valves 50 mm and smaller shall be bronze with screwed connections. Valves 60 mm and larger shall be ironed body, bronze trim, solid disc, and flanged

C. Alarm Check Valves: 175 psi WWP, approved type with main drain valve, pressure gauges to indicate pressure above and below the seat, side outlet globe valves for testing, alarm control, and test cocks, restriction bypass, retarding chamber with drip pipe, drip receiver with removable cover and pressure alarm switch.

D. Tamper Switches: Switch shall close contacts when valve tampering occurs. Coordinate electrical characteristics with Fire Alarm System.

E. Deluge Valves: Ductile-iron body, 175 psig working pressure, hydraulically operated, differential-pressure-type valve. Valves shall have flanged inlet and outlet, and bronze seat with O-ring seals. Include trim sets for bypass, drain, electric sprinkler alarm switch, pressure gages, drip cup assembly piped without valves separate from main drain line, fill line attachment with strainer, and push rod chamber supply connection.
   1. Option: Grooved-end connections for use with grooved-end piping.
   2. Dry pilot line trim set including dry pilot actuator, air and water pressure gages, low air pressure warning switch, air relief valve, and actuation device. Dry pilot line actuator includes ductile iron,

175 psig working pressure, air operated, diaphragm-type valve with resilient facing plate, resilient diaphragm, and replaceable bronze seat. Valve includes threaded water and air inlets and water outlet. Loss of air pressure on dry pilot line side allows pilot line actuator to open and causes deluge valve to open immediately.

3. Thermostatic Releases: Corrosion resistant rate-of rise releasing device for use on pneumatic release systems controlling operation of deluge valves.
4. Manual Control Stations: Hydraulic operation, with union, 1/2-inch pipe nipple, and bronze ball valve. Include metal enclosure labeled “MANUAL CONTROL STATION” with operating instructions and a cover held closed by breakable strut to prevent accidental opening. Cover must be replaced after each opening.

F. Zone Control Valves: Provide zone control valves as indicated on the sprinkler riser diagram. Each zone control station shall be provided with a supervised shut off valve, flow switch, drain valve, and downstream pressure gauge.

2.7 PIPE AND FITTINGS

A. Pipe and Connections: Pipe shall be eddy current or hydrostatic tested, black painted steel, Schedule 40 ASTM A 135 and ASTM A 795 with welded, threaded, rolled or cut grooved end connections for sizes 6" and smaller, Schedule 30 ASTM A 135 or A 53 with welded, threaded, rolled or cut grooved end connections for sizes 8" and larger, designed for 175 psi working pressure, and in accordance with NFPA 13. Fittings shall be black malleable or ductile iron.

B. Grooved Couplings: Victaulic Zeroflex Type consisting of ASTM A 47 malleable iron housings, ASTM D 2000 gaskets, ASTM A 183 carbon steel nuts and bolts. Provide matching grooved or shouldered fittings and pipe ends, designed to accept coupling assemblies. Outlet couplings, plain end fittings, mechanical tee assemblies, and hooker fittings shall not be used. NOTE: Over the floor plan area of the fab, no grooved fittings shall be allowed.

C. Provide grooved flexible type couplings where required by FM Data Sheet 2-8N or NFPA 13, whichever is more stringent.

D. Paint all fire protection in clean areas or exposed piping with approved type paint in accordance with

Section 09900, Painting Equipment and Piping, color to be red.

E. Sprinkler drain piping shall be metallic in accordance with NFPA 13.

2.8 HANGERS

A. Provide pipe hangers and building attachments in accordance with NFPA Standard No. 13, and Section 15240-A, Mechanical Vibration Control, with requirements for use in sprinkler system.
   1. The use of powder driven devices is not acceptable.

2.9 PORTABLE EXTINGUISHERS

A. Portable extinguishers shall be provided in accordance with Local code. General locations, types, and sizes shall be as indicated.

B. For general use, combination ABC dry chemical extinguishers shall be furnished in one size only: 9kg units rated 4-A:40-B:C pressure stored type. Valves shall be metal. Plastic construction will not be allowed.
   1. Type ABC extinguishers shall not be used in electrical rrooms, cleanrooms, or HPM rooms.

C. Carbon dioxide extinguishers shall be capable of extinguishing class A, B, and C type fires. Extinguishers shall be supplied with horn, hose, and valve assembly. The discharge of the gas shall be through a suitable nozzle and regulated a valve of the trigger or thumb-push type.
   1. Portable carbon dioxide fire extinguisher containers shall lbe made of lightweight material.
   2. Wheeled carbon dioxide extinguishers shall be provided for each floor of the CUP, cleanroom, clean subfab, and utility subfab.

D. Portable fire extinguishers shall be shop tested in accordance with applicable UL and Taiwan standards.

E. Extinguisher ratings and performance during fire tests shall be in accordance with UL 711.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Sprinkler heads shall be positioned at centers of acoustic ceiling tiles and as required for an orderly, symmetrical appearance throughout the project, unless specifically indicated otherwise on reflected ceiling plans.

B. Coordinate sprinkler piping routing with ductwork, pipes, structure, conduit, and other building components. Provide auxiliary drains, pipe offsets required to clear other work. Relocate sprinkler piping which has not been coordinated with other work at no cost to Owner.

C. Arrange, phase, and perform work to assure adequate services for the Owner at all times.

D. Run piping concealed above furred ceilings and in joists to minimize obstructions. Expose only heads.

E. Protect sprinkler heads against mechanical injury with standard guards.

F. Provide tamper switches for each sprinkler control valve. Switches shall be connected to fire alarm system under Division 16.

G. Provide flow switches, which close contact when flow is detected. Adjust delay to 30 seconds. Install flow switches and adjacent valves to be easily accessible and behind access panels when located above plaster ceilings. Locate flow switch minimum of 30 cm from a fitting that changes the direction of flow and not less than 60 cm from a drain connection or 10 pipe diameters from a gate, check, or alarm valve. Switches will be connected to fire alarm system under the work of Division 16.

H. Provide drains at base of risers, on all valved sections, and at other locations for complete drainage of system. Route drain to outside splash block at a location which will accept full flow under full pressure without damage to surrounding landscaping or buildings.

I. Attach inspector’s test connection, at remote portion of system, where indicated. Route discharge to outside. Locate test valve in a readily accessible location.

J. Flush system before connecting sprinkler to underground supply connection in accordance with requirements of NFPA Standard No. 13 and FM Data Sheets 2-8N and 3-10.

K. Provide spacing and sizing of pipe hangers per FM Data Sheet 2-8N.

L. Provide seismic bracing for fire protection piping systems. Provide design and components per FM Data Sheet 2-8N and NFPA 13.

M. For hose stations in the Fabrication Building clean room, and where static pressure exceeds 100 psi at hose station, provide pressure reducing valve to prevent pressure on hose from exceeding 100 psi.

N. Install fire hose cabinets with cabinet top maximum 2.1 m above floor. Locate angle valve in cabinet maximum 0.3 to 1.5 m above floor.

3.2 INSTRUCTIONS

A. After the instructions have been submitted and approved, furnish the Engineer and Owner with 1 bound copy each, of complete instructions, including catalog cuts, diagrams, Drawings, and other descriptive data covering the proper testing, operation, and maintenance of each type of system installed, and the necessary information for ordering replacement parts. In addition, post one copy of complete instructions at alarm valve locations, in metal frame with rigid plastic cover.

B. Provide Owner’s maintenance personnel with detailed instructions covering the necessary and recommended testing, operating, and maintenance procedures for each type of system.

3.3 TESTS

A. Upon completion and prior to acceptance of the installation, subject the system, including the underground supply connection, to all tests per FM Data Sheets 2-8N and 3-10; NFPA Standard No. 13, and tests required by the Local authorities. Furnish the Engineer and Owner with a certificate thereof.

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